CHEM 1210
Quiz 1
Given Time: 1 hr.

Name: _____ Student ID: _____

**Multiple choice. (50 pts. 2 pts each). Circle the correct answer.**

1. Which of the following represents a *hypothesis*?
   A. When a substance combusts, it combines with air.
   B. Sodium reacts with water to form sodium hydroxide and hydrogen gas.
   C. Argon gas is a fairly inert substance.
   D. Water is a liquid at room temperature.
   E. When paper burns, heat is given off.
2. Determine the volume of an object that has a mass of 455.6 g and a density of $19.3 g/cm^3$.
   A. 87.9 mL
   B. 23.6 mL
   C. 42.4 mL
   D. 18.5 mL
   E. 31.2 mL
3. *Systematic error* is defined as
   A. error that has equal probability of being too high and too low.
   B. error that averages out with repeated trials.
   C. error that tends to be too high or too low.
   D. error that is random.
4. What answer should be reported, with the correct number of significant figures, for the following calculation? (433.621 - 333.9) × 11.900
   A. $1.187 \times 10^3$
   B. $1.1868 \times 10^3$
   C. $1.18680 \times 10^3$
   D. $1.19 \times 10^3$
   E. $1.186799 \times 10^3$
5. The density of air under ordinary conditions at 25°C is 1.19 g/L. How many kilograms of air are in a room that measures 11.0 ft × 12.0 ft and has a(n) 10.0 ft ceiling? 1 in. = 2.54 cm (exactly); $1 L = 10^3 cm^3$
   A. 3.99
   B. 0.166
   C. $4.45 \times 10^4$
   D. 0.1145
   E. 44.5

6. A student performs an experiment to determine the density of a sugar solution. She obtains the following results: 1.80 g/mL, 1.79 g/mL, 1.80 g/mL, 1.79 g/mL. If the actual value for the density of the sugar solution is 1.79 g/mL, which statement below best describes her results?
   A. Her results are both precise and accurate
   B. Her results are precise, but not accurate.
   C. Her results are accurate, but not precise.
   D. Her results are neither precise nor accurate.
   E. It isn't possible to determine with the information given.
7. How many significant figures are in the measurement $1.04 \times 10^{-4}$ g?
   A. 1
   B. 3
   C. 2
   D. 4
   E. 5
8. If a solution has a temperature of 55 K, what is its temperature in degrees Celsius?
   A. 131°C
   B. 328°C
   C. -218°C
   D. 12.8°C
   E. 155°C
9. Which of the following is an example of the law of multiple proportions?
   A. A sample of chlorine is found to contain three times as much Cl-35 as Cl-37.
   B. Two different samples of table salt are found to have the same ratio of sodium to chlorine.
   C. The atomic mass of bromine is found to be 79.90 amu.
   D. Two different compounds formed from carbon and oxygen have the following mass ratios: 1.33 g O: 1 g C and 2.66 g O: 1 g C.
   E. Nitrogen dioxide always has a mass ratio of 2.28 g O: 1 g N.

10. Determine the number of protons, neutrons, and electrons in the following: $^{40}_{18}X$

A. $p^+ = 18$ $n° = 18$ $e^- = 22$

B. $p^+ = 22$ $n° = 18$ $e^- = 18$

C. $p^+ = 18$ $n° = 22$ $e^- = 40$

D. $p^+ = 40$ $n° = 22$ $e^- = 18$

E. $p^+ = 18$ $n° = 22$ $e^- = 18$

11. Calculate the atomic mass of silver if silver has 2 naturally occurring isotopes with the following masses and natural abundances:

| | | |
|---|---|---|
| Ag-107 | 106.90509 amu | 51.84% |
| Ag-109 | 108.90476 amu | 48.46% |

A. 108.19 amu

B. 107.90 amu

C. 108.00 amu

D. 107.79 amu

E. 108.32 amu

12. How many moles of Kr are contained in 398 mg of Kr?

A. 33.4 moles Kr

B. $4.75 \times 10^{-3}$ moles Kr

C. $2.11 \times 10^{-4}$ moles Kr

D. $2.99 \times 10^{-3}$ moles Kr

E. $1.19 \times 10^{-4}$ moles Kr

13. In which of the following sets do all species have the same number of electrons?

A. C, $N^{3-}$, $O^{2-}$

B. $Mg^{2+}$, $Sr^{2+}$, $Ba^{2+}$

C. $Cl^-$, Ar, $Ca^{2+}$

D. Br, $Br^-$, $Br^+$

14. How many xenon atoms are contained in 2.36 moles of xenon?

A. $3.92 \times 10^{24}$ xenon atoms

B. $2.55 \times 10^{23}$ xenon atoms

C. $7.91 \times 10^{25}$ xenon atoms

D. $1.42 \times 10^{24}$ xenon atoms

E. $1.87 \times 10^{26}$ xenon atoms

15. Give a possible molecular formula for the empirical formula of $C_2H_5N$.

A. $C_4H_{10}N$

B. $C_5H_{10}N_2$

C. $C_6H_{15}N$

D. $C_4H_5N_2$

E. $C_4H_{10}N_2$

16. Give the formula for sulfurous acid.

A. $H_2SO_3$

B. $HSO_3$

C. $H_2SO_4$

D. $HSO_4$

E. $H_2SO_2$

17. Calculate the molar mass of $Al(C_2H_3O_2)_3$.

A. 86.03 g/mol

B. 204.13 g/mol

C. 56.00 g/mol

D. 258.09 g/mol

E. 139.99 g/mol

18. Calculate the mass percent composition of sulfur in $Al_2(SO_4)_3$.

A. 9.372%

B. 42.73%

C. 28.12%

D. 21.38%

E. 35.97%

19. Determine the molecular formula of a compound that has a molar mass of 92.0 g/mol and an empirical formula of $NO_2$.

    A. $N_2O_3$
    B. $N_3O_6$
    C. $NO_2$
    D. $N_2O_4$
    E. $N_2O_5$

20. Write a **balanced** equation to show the reaction of sulfurous acid with lithium hydroxide to form water and lithium sulfite.

    A. $H_2SO_4(aq) + LiOH(aq) \rightarrow H_2O(l) + Li_2SO_4(aq)$
    B. $HSO_3(aq) + LiOH(aq) \rightarrow H_2O(l) + LiSO_3(aq)$
    C. $HSO_4(aq) + LiOH(aq) \rightarrow H_2O(l) + LiSO_4(aq)$
    D. $H_2S(aq) + 2\ LiOH(aq) \rightarrow 2\ H_2O(l) + Li_2S(aq)$
    E. $H_2SO_3(aq) + 2\ LiOH(aq) \rightarrow 2\ H_2O(l) + Li_2SO_3(aq)$

21. How many grams of $Li_3N$ can be formed from 1.75 moles of Li? Assume an excess of nitrogen.

$$6Li(s) + N_2(g) \rightarrow 2\ Li_3N(s)$$

    A. 20.3 g $Li_3N$
    B. 18.3 g $Li_3N$
    C. 58.3 g $Li_3N$
    D. 61.0 g $Li_3N$
    E. 15.1 g $Li_3N$

22. A 12.39 g sample of phosphorus reacts with 42.54 g of chlorine to form only phosphorus trichloride ($PCl_3$). If it is the only product, what mass of $PCl_3$ is formed?

    A. 30.15 g
    B. 54.93 g
    C. 140.01 g
    D. 79.71 g
    E. 91.86 g

23. According to the following reaction, what amount of $Al_2S_3$ remains when 20.00 g of $Al_2S_3$ and 2.00 g of $H_2O$ are reacted? A few of the molar masses are as follows: $Al_2S_3$ = 150.17 g/mol, $H_2O$ = 18.02 g/mol.

$$Al_2S_3(s) + 6\ H_2O(l) \rightarrow 2\ Al(OH)_3(s) + 3\ H_2S(g)$$

A. 28.33 g
B. 14.00 g
C. 17.22 g
D. 8.33 g
E. 19.78 g

24. Given the initial amounts of reactants, find the limiting reactant and determine how much product is formed (in g) for the following reaction:

$$4\ Al\ (s) + 3\ O_2\ (g) \rightarrow 2\ Al_2O_3\ (s)$$

13.2 mol Al; 12.6 mol $O_2$

A. LR = $O_2$; 856 g $Al_2O_3$
B. LR = Al; $2.7 \times 10^3$ g $Al_2O_3$
C. LR = $O_2$; $1.9 \times 10^3$ g $Al_2O_3$
D. LR = Al; 673 g $Al_2O_3$
E. None of these are the limiting reactant.

25. Identify the combustion reaction.

A. $2\ C_3H_7OH(l) + 9\ O_2(g) \rightarrow 6\ CO_2(g) + 8\ H_2O(g)$
B. $2\ H_2(g) + O_2(g) \rightarrow 2\ H_2O(g)$
C. $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O$
D. $C(s) + O_2(g) \rightarrow CO_2(g)$
E. All of the reactions are examples of combustion reactions.

Name: _____ Student ID: _____

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